Are meteorological, agricultural, and satellite features from the CY-Bench Dataset good predictors of European Wheat Futures Price Changes?

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1 Executive Summary

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There are two sets of research topics that motivate our literature review: the application of machine learning to multi-national crop yield forecasting, and the use of data-driven models to forecast wheat futures using agricultural signals as inputs. Both areas of research have developed substantially in recent years, but largely in parallel, with limited work connecting yield forecasting outputs to commodity price models. The review below covers both sets of papers.

We start with the foundations of ML-based crop yield prediction as described by van Klompenburg, Kassahun, and Catal [52], who conducted a systematic literature review of 50 studies across a range of crops, algorithms, and geographies. The feature sets and model choices most commonly adopted were temperature, rainfall, and soil type as the dominant predictors. In terms of model selection, their review found that no single model emerged as definitively best, but that Random Forest, Neural Networks, Linear Regression, and Gradient Boosting were the most frequently used, with most studies testing a variety of models to identify the best performer for their specific task. The most widely used evaluation method was 10-fold cross-validation, and artificial neural networks were the most widely applied model class. They also looked at deep learning literature and found CNNs, LSTMs, and DNNs to be the core architectures for this prediction task. Importantly, their review highlighted data availability as the primary constraint on generalization. They found that studies were typically operated over limited geographic extents with short time series, making it difficult to draw conclusions about which methods transfer across regions or climates. Oikonomidis, Catal, and Kassahun [35] extended the ML crop yield review specifically to deep learning approaches, systematically analysing 44 primary studies. Their results confirmed CNN as both the most frequently applied deep learning algorithm and the best-performing one across studies, with LSTM and DNN following as the next most common architectures. They found that RMSE was the dominant evaluation metric used across the reviewed literature. Crucially, they also found that the most significant challenge was the limitation of large training datasets due to insufficient data availability. This raised the risk of overfitting and reduced real-world model performance, with proposed solutions including data augmentation and transfer learning. This data constraint is consistent with what van Klompenburg et al. [52] identified in their ML literature as the primary constraint, so together the two reviews establish data availability and geographic coverage as the primary bottleneck to building generalisable crop yield prediction models. Muruganantham et al. [34] conducted a further systematic review specifically combining deep learning with remote sensing for crop yield prediction, analysing 44 studies from 2012 to 2022 and confirming the previous findings that LSTM and CNN are the dominant approaches, with satellite-derived vegetation indices emerging as among the most informative input features alongside weather variables.

Paudel et al. [36] attempt to address these gaps by establishing a large-scale ML baseline for crop yield prediction across multiple countries and crop types. Working with a feature set covering meteorological variables, crop model outputs, and remote sensing indicators, they evaluated Ridge Regression, KNN, SVM, and Gradient Boosting under a time-based look-forward cross-validation scheme that simulates operational forecasting conditions by training only on data preceding each test year on 70 percent train, 30 percent test split. They looked at three European countries, the Netherlands, Germany, and France, evaluating five crop types, with predictions made at the regional level and then aggregated to the national level for comparison with the established European Commission's "Monitoring Agricultural ResourceS Crop Yield Forecasting System" (MCYFS). Their results showed that early-season predictions made 30 days after planting were broadly competitive with the MCYFS in most cases. They note that accuracy improved as the season progressed toward harvest, confirming that lead time is a key constraint for any real-time application.

The most recent CY-Bench paper Paudel et al. [37] built on this foundation at a global scale, combining predictor data, including meteorological variables such as temperature, precipitation, evapotranspiration, and solar radiation, with soil data like available water capacity, bulk density, drainage class, and moisture content. Lastly, they also used satellite vegetation across 29 wheat-growing and 42 maize-growing countries. For features, the dataset includes threshold-based indicators such as counts of cold days, hot days, and dry days during the growing season, for the purpose of tracking climate-related stress signals on the crops. Model evaluation uses Leave One Year Out cross-validation to obtain estimates of performance across both typical and extreme harvest years, with normalized root mean squared error and mean absolute percentage error as the primary metrics to support cross-country comparison. Baseline models evaluated include Ridge Regression, Random Forest, and LSTM, which were all compared against a Naive Average Yield benchmark. These two Paudel studies together define the current benchmark results for the CY-Bench dataset for crop yield forecasting.

In terms of modelling agricultural commodity price dynamics, Theofilou et al. [50] provide a systematic review of 50 machine learning studies applied to wheat, corn, and rice price forecasting. Their finding is that hybrid deep learning models consistently outperform traditional time series approaches such as ARIMA and classical ML approaches of PLS Regression and Neural Networks, while still identifying persistent challenges including reliance on region-specific datasets and limited model interpretability. This broad finding motivates the more specific studies that follow, each of which deals with the core difficulty of forecasting a volatile and structurally complex price series. Li [29] illustrates this directly by benchmarking Holt-Winters exponential smoothing against three LSTM-based variants on a European wheat spot price series, finding that the CNN-LSTM model best captures long-term dependencies and price trend characteristics, confirming that purely parametric smoothing methods are insufficient for a commodity as volatile as wheat. Zhang and Tang [54] take this a step further by targeting agricultural commodity futures prices specifically, proposing a hybrid LSTM model that first decomposes the price series using variational mode decomposition before prediction. Their key finding is that this decomposition in the preprocessing steps substantially improves accuracy for volatile futures series, suggesting that the structural complexity and fast moving nature of futures prices require dedicated handling beyond what standard LSTM architectures provide. Brignoli et al. [6] reinforce this conclusion in the grain futures context, finding that LSTM-RNNs consistently outperform classical econometric models at longer forecast horizons precisely because of their ability to handle sudden structural breaks typical of commodity market movement. However, they also note that LSTM models struggle with seasonality and trend components, requiring explicit pre-processing steps for their removal, a similar finding to Zhang et al[54] findings. Taken together, these studies establish that deep learning and hybrid approaches represent the state of the art for agricultural price forecasting, but that all of them rely exclusively on price history and market signals as inputs. Thaker, Chan, and Sonner [49] take a fundamentally different approach that bridges the yield and price literatures directly since they apply a CNN to aerial imagery of winter hard red wheat planted areas. They use cloud coverage patterns as a weather proxy for field-level yield conditions, and then train the model on this imagery to produce a 20 trading-day forecast for wheat futures price on the Chicago Mercantile Exchange with purely directional long or short trading recommendations. Their paper demonstrates that satellite-derived signals can generate positive alpha in wheat futures, but they also document performance decay once the signal becomes widely available. This is, again, a natural and expected limitation of this type of prediction task, potentially showing that the informational advantage of agricultural based price forecasts depend on signal differentiation from other traders.

Together, the reviewed literature establishes a clear gap. Indeed, the yield prediction task has produced well-validated, multi-national, in-season yield forecasts but has not connected these directly to commodity futures price modelling. Secondly, the literature on the price modelling task demonstrates the growing effectiveness of deep learning and hybrid models for agricultural price forecasting, but these approaches tend to focus on price history and market signals rather than on benchmarked yield predictors. Finally, Thaker et al. [49] come closest to bridging this gap by using satellite-derived yield proxies as price signals, however, they mainly rely on unvalidated aerial imagery rather than more established predictors. Our study attempts to address this gap by using CY-Bench yield predictors as structured fundamental inputs to a wheat futures price model.

4 Data Preparation

This next section describes how we prepare the CY-Bench dataset for two distinct prediction tasks, namely, European Wheat Futures Prediction and Yield Prediction prior to harvest. We start by describing the preparation steps shared by both prediction tasks, namely the agricultural data and the daily harmonisation via last-observation-carried-forward interpolation. Secondly, we describe the wheat futures roll-adjusted data preparation justifying our key choices. Thirdly, we cover the two distinct region filtering choices we applied to the futures price prediction dataset as well as the yield prediction dataset. Finally, we obtain two datasets from the same underlying CY-Bench source: a daily-frequency panel for the wheat futures return prediction task, and an annual cross-section for the crop yield forecasting task.

4.1 Agricultural Data and Daily Harmonisation

Firstly, our agricultural data is taken from the CY-Bench dataset [37], which provides sub-national time series of climate, soil, and vegetation variables for 29 wheat growing countries (although more crops are covered too). We restrict our analysis to France, Germany, and Poland totalling over 500 regions. We select these three countries for two reasons. First, they are the three largest wheat-producing EU member states, collectively accounting for approximately 43% of total EU wheat output [14] and they are, therefore, the countries whose supply conditions are most likely to influence the Euronext Milling Wheat contract that serves as our prediction target. Second, France and Germany appear in the Paudel et al. [36] benchmark study, providing established yield forecasting results against which we can validate our secondary task. For each region, CY-Bench has five categories of predictor variables which are kept at their native temporal resolutions. Meteorological variables from the AgERA5 reanalysis dataset [5] include daily minimum and maximum temperature (°C), precipitation flux (mm), and solar radiation flux ($\mathrm{J\,m^{-2}}$). Further, we have Soil moisture indicators from the NASA Global Land Data Assimilation System (GLDAS) [41] provide daily surface and root-zone soil moisture ($\mathrm{kg\,m^{-2}}$), these capture the water available to crops below the surface. Additionally, Satellite-derived vegetation indices comprise two variables that measure crop health from space. The Normalised Difference Vegetation Index (NDVI), sourced from the MODIS sensor at 8-day resolution, quantifies how green and photosynthetically active the land surface is by comparing the reflectance of visible red light (which healthy vegetation absorbs) with near-infrared light (which it reflects); values range from -1 to 1, with higher values indicating denser, healthier vegetation [51]. The second satellite derived indice is the Fraction of Absorbed Photosynthetically Active Radiation (fPAR), sourced from the JRC at dekadal (approximately 10-day) resolution, measures the share of incoming sunlight in the wavelengths used for photosynthesis (400-700 nm) that is actually absorbed by the crop canopy, expressed as a percentage [47]. NDVI is an indirect proxy for vegetation vigour based on spectral reflectance and fPAR directly estimates how much solar energy the crop is capturing for growth, this makes the two variables complementary indicators of crop condition by nature.

We then create our features from these climate variables which all have different temporal resolutions we must harmonise the data so as to avoid data leakage. Here we choose to simply retain these frequencies and expand them to daily increments per region using last-observation-carried-forward (LOCF) interpolation, i.e. we fill each day with the most recently past observed value. LOCF is a standard approach in time series data imputation, however, by assuming the data is constant, it evidently introduces bias. Here, we specifically choose this method to simulate a real operational context in which we may not have access to the variables' true value. After this procedure, each region has a consistent set of variables aligned to a common daily calendar.

Finally, we note that the agricultural data's availability starts at different periods for different features, for instance, meteorological data starts in 01-01-2001 while soil data start 01-02-2003. Thus, we decide to start our dataset around when GLDAS soil moisture data becomes available [37] in 01-01-2003 to ensure full data availability. Further, we ensure that our dataset finishes at the last observation for which all data agricultural variables are available, fPAR's last data point is on the 21-12-2023 which is the earliest of all variables, thus, our dataset ends in 2023. Further details are later discussed in Section 6 for the scope of the yield prediction dataset.

4.2 Futures Price Data from Bloomberg

Secondly, our target variable data for the futures price prediction task is the Euronext Milling Wheat roll-adjusted daily closing prices obtained from Bloomberg for the period 01-01-2003 to 31-12-2023 (to align with our agricultural data), yielding 5,389 trading days. More precisely, we define the raw target data as the closing price of the earliest maturing Euronext Milling Wheat Futures Contract. This is because near-term futures contracts are usually most liquid as per Ripple and Moosa [42]. We use Bloomberg's ratio roll-adjustment to the time series 10 days before each contract expiry, this means that historical prices are multiplied by $P_{new\ contract}/P_{old\ contract}$ so as to preserve the percentage returns across all rolls. Our choice of 10 days before expiry is due to liquidity factors, indeed, futures contract generally decrease in liquidity as they get closer to maturity as per Ding and Lim [13]. Further, Carchano and Pardo's [8] study of stock index futures shows that choice of rollover timing does not induce statistically significant differences between the resulting return series. However, their study focuses on stock index futures rather than agricultural commodities which may exhibit stronger maturity effects due to seasonal supply factors. As per López de Prado Chapter 2 [27], roll-adjustment is a key step as it eliminates the artificial jumps that occur at expiration to preserve the return dynamics experienced by a roll-following investor. This property is directly relevant to our study because the prediction target is a percentage return, not an absolute price level. A limitation of ratio adjustment is that it distorts historical price *levels* (early-period prices may become very small or very large), making the series unsuitable for level-based analysis; however, since we use only returns and return-derived features, this limitation does not affect our study.

We retain two key fields from the raw data, closing price as well as volume. We note that while price has no missing data, missing volume data is forward-filled using LOCF. In this case, the missing volume data represents only 290 trading days or c.5.4% of trading days.

4.3 Region filtering and excluded areas

Lastly, we detail our region filtering steps for the futures price prediction task where we exclude urban/non-wheat producing regions and then for the yield prediction task where we exclude regions with significant missing yield data due to lack of reporting.

Looking at the futures prediction task, When creating our agricultural features we exclude some highly urban and non-wheat producing regions as they should not directly be influencing the supply of wheat. We implement this simply filtering regions that are not included within CY-bench's raw yield data. Indeed, as per our Table A1 in the Appendix, we see that these regions are either urban, logistic hubs or non-wheat producing. This means that we retain 507 regions across France (94 regions), Germany (396 regions), and Poland (17 regions) (out of 518).

For the yield forecasting task, we apply a different region filter. Indeed, we retain only the 357 regions (90 French, 251 German, 16 Polish) that report wheat yield statistics for every year in the 2003 to 2020 window. The start year is the same as the futures prediction task dataset i.e. the start of the GLDAS soil moisture data. However, the end year is constrained by the latest available sub-national yield statistics from the JRC harmonised European dataset [43], which provides French and Polish NUTS-level yields through 2020. Although these filters lead to some information loss, it is not material. The 357 retained regions account for 97.5% of total wheat production across the three countries (99.9% for France, 93.3% for Germany, 99.4% for Poland). Indeed, the dropped regions are predominantly small districts with discontinued reporting rather than major wheat-producing areas. This ensures that we have complete temporal coverage for every retained region for fair cross-regional comparison under the walk-forward evaluation scheme described in Section 7.

5 Exploratory Data Analysis

In this next section, we cover our Exploratory Data Analysis (EDA) which drives several of our choices. Firstly, we cover how the distribution of returns and volatility lead us to implementation a regression task with a specific train-test split around 2017. Secondly, we look at the timescale mismatch between agricultural variables and returns. Thirdly, we look at the correlation between features. Finally, we focus on the yield prediction task. We note that the following analysis is computed over the full sample period and not only the training set. However, none of our observations depend on specific test-period data or statistics. Indeed, our modelling decisions are informed by properties of the data which are stable across the training and test windows, thus, there is no information leakage in our modelling decisions.

5.1 Futures Returns Distribution and Volatility

First, we show the distribution and volatility of our prediction target for the futures task which is the 5-trading-day forward return $y_t = P_{t+5}/P_t - 1$, where P_t is the roll-adjusted closing price (more details in Section 6). As shown in Figure 1(a), the distribution of the target is approximately symmetric with skewness of 0.90 and centred near zero mean = 0.18%, but exhibits heavy tails with an excess kurtosis of 6.17. This is consistent with the well-documented leptokurtosis of commodity futures returns [31]. To quantify the tail behaviour, we fit a Student-t distribution to the returns and obtain 3.6 degrees of freedom (see Figure B2 in the Appendix), confirming that the tails are substantially fatter than Gaussian meaning that extreme weekly moves are more frequent than a normal model would predict. In practice, 1.1% of observations exceed ±10% in absolute value. Figure 1(b) shows that return volatility is not constant over time: the 2007-08 food crisis and the 2022 Russia-Ukraine war both produced sustained volatility roughly two to three times higher than the 2012-2016 baseline. Our observations lead us to define the futures prediction task as a regression problem, i.e. predicting continuous returns rather than a binary classification of direction, since the return magnitudes vary significantly and are economically meaningful. Further, for the futures prediction task, we place the chronological train/test boundary so that the training set includes both the 2007-08 and 2010-11 volatile episodes, ensuring the model has seen extreme market conditions before encountering the COVID-19 and Ukraine shocks in the test period (the yield prediction task has a different split given the nature of the task).

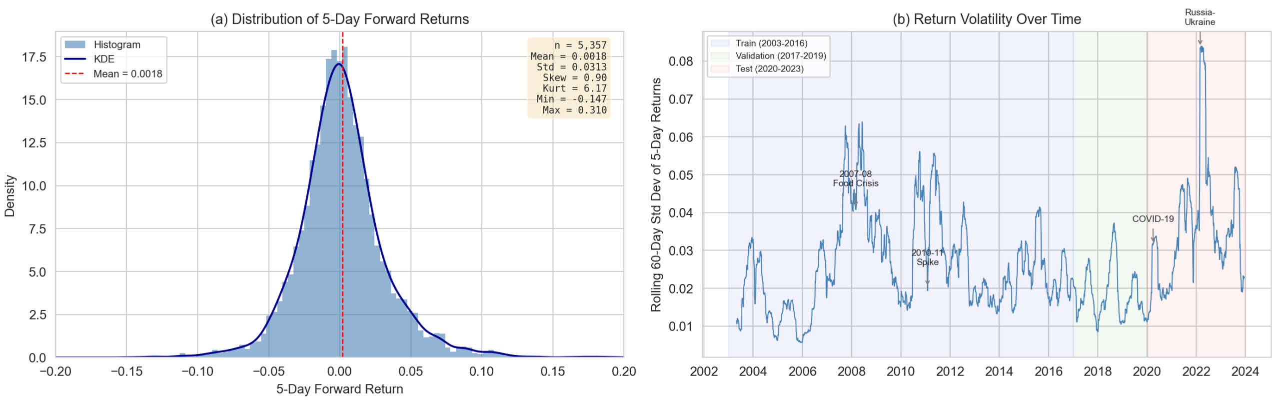

Fig. 1 (a) Distribution of 5-day forward returns for the Euronext Milling Wheat futures contract (2003-2023). The distribution is approximately symmetric with heavy tails (kurtosis = 6.17). (b) Rolling 60-day standard deviation of 5-day returns, with train/validation/test periods shaded. Volatility spikes correspond to the 2007-08 food crisis, COVID-19, and the 2022 Russia-Ukraine war.

5.2 Agricultural Feature Behaviour

We now look at Figure 2 which plots the Euronext wheat futures 5-trading day returns alongside France-average NDVI and 30-day cumulative precipitation over the full sample period. We immediately see that returns are noisy and do not exhibit a trend while agricultural data is smooth and seasonal. Indeed, the agricultural variables exhibit strong annual seasonality: NDVI peaks each June-July when the wheat harvest areas reach maximum greenness, then declines after harvest, this repeats with high regularity across all twenty years. Figure B1 in Appendix shows that cumulative precipitation fluctuates on a similar seasonal cycle but with much greater variability. In contrast, the futures return series shows no consistent seasonal pattern and can often be driven by geopolitical and macroeconomic events. Further, as mentioned in section 4, agricultural data is smooth on a daily timescale with NDVI updating every 8 days and fPAR every 10 days. This timescale mismatch is a key motivation for our choice of a 5-day (one-week) forecast horizon rather than a daily one: the slow-moving agricultural signals are unlikely to predict daily price variation, which is dominated by microstructure noise and order-flow dynamics [11] (See Section 6 for more details). Additional EDA figures in Appendix B show that NDVI and fPAR are highly correlated ($\rho = 0.93$), however, they potentially diverge during cloudy periods (as NDVI is derived from satellite imagery), thus, we retain both in our feature set. Further, surface and root-zone soil moisture capture different timescales of water availability, thus, we keep both despite their near-perfect correlation of $\rho = 0.99$. Further, we see that the 2018 European drought produced 2.5 times the average number of hot stress days in Germany during the growing season, this drives our threshold-based stress features in Section 6.

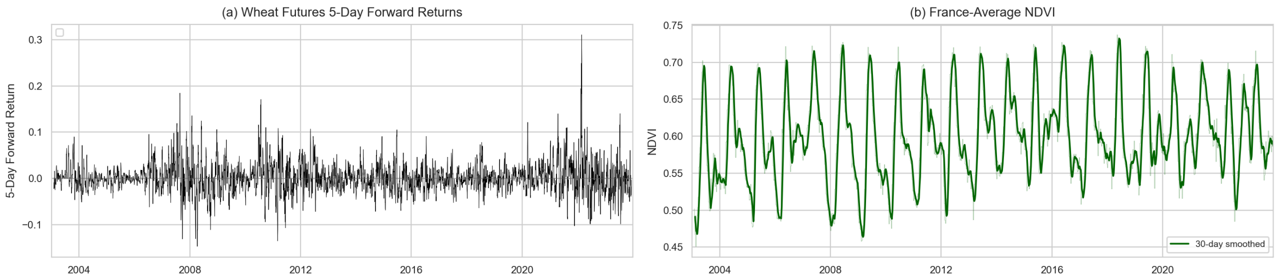

Fig. 2 (a) 5-day forward returns of the Euronext Milling Wheat futures, returns are noisy and display no seasonal pattern. (b) France-average NDVI with 30-day smoothing. NDVI exhibits strong annual seasonality, peaking each June-July. The timescale mismatch between the two series motivates the choice of a weekly rather than daily forecast horizon.

5.3 Feature Correlations

Figure B8 in Appendix B presents the Fisher z-averaged correlation matrix across all 507 regions for the 12 raw CY-Bench variables. Of the 66 unique feature pairs, 9 (13.6%) exceed an absolute correlation of 0.85, with the highest being $|\rho| = 0.98$ between tmin and tavg. The most correlated group of features involves the temperature-related variables: tmin, tmax, tavg, et0, and vpd, which are all correlated above 0.80. This is expected since evapotranspiration and vapour pressure deficit are both derived from temperature inputs. As previously mentioned, surface and root-zone soil moisture are also highly correlated ($\rho = 0.97$). Precipitation stands out as largely uncorrelated with most other variables ($|\rho| < 0.33$), however, its correlation with the climatic water balance (cwb) is very high ($\rho = 0.94$) since cwb is computed as precipitation minus evapotranspiration. Although these patterns would introduce multicollinearity, we decide to keep all of these variables given that they represent distinct physical processes which may bring additional information to our models. We still verify the correlation of features during our feature engineering step in Section 6.

5.4 Yield Distribution Across Countries

Finally, we briefly examine the yield data for our secondary prediction task. Figure 3(a) in Appendix B shows that average wheat yields differ substantially across the three countries: Germany at 7.35 t/ha, France at 6.17 t/ha and Poland at 4.12 t/ha, reflecting differences in climate, soil quality, and farming intensity. Germany also exhibits the widest absolute spread across its 251 NUTS3 regions, with yields ranging from 2.57 to 11.04 t/ha. Figure 3(b) plots the average yield trajectory over time. All three countries display a slight upward trend which could be driven by long-term improvements in crop varieties and management practices. Additionally, we see that, in 2018, all three countries experience a dip which corresponds to the European drought: Germany experienced a 7.3% decline relative to other years, Poland 5.6%, and France 2.9%. These cross-country differences in yield level, variability, and drought sensitivity motivate our decision to tune model hyperparameters separately per country for the yield prediction task rather than using a single EU-wide configuration, as discussed in Section 7.

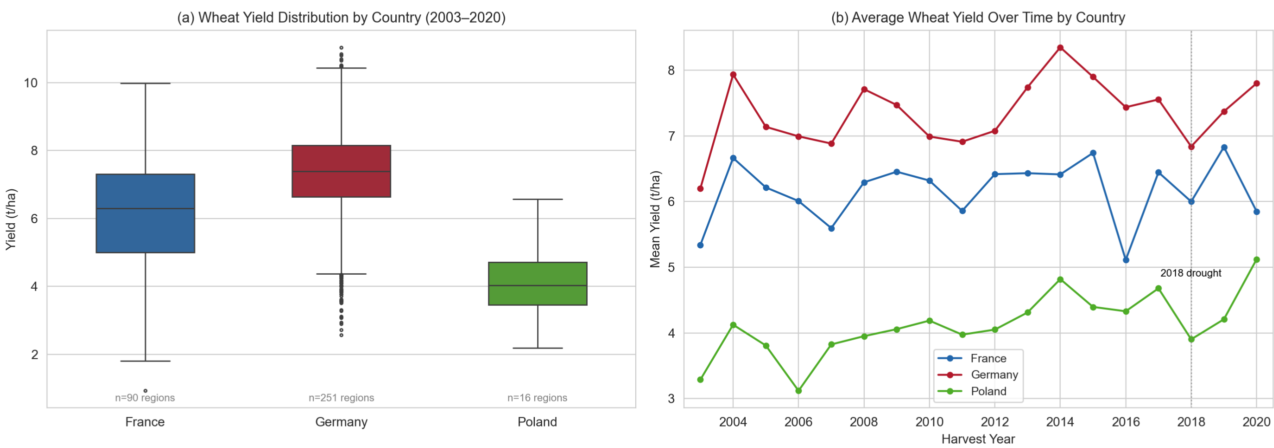

Fig. 3 (a) Wheat yield distributions by country (2003-2020). Germany has the highest average yield (7.35 t/ha) and the widest spread across its 251 NUTS3 regions. (b) Average wheat yield over time by country. The 2018 European drought is visible as a synchronised dip, with Germany experiencing the largest decline (−7.3% relative to other years).

6 Feature Engineering and Prediction Design

Now that we have pre-processed and visualised our data, we add features and transform the data for model fitting for both prediction tasks. We start by describing the 14 engineered features for the futures prediction task, then we detail futures prediction target construction and temporal alignment, and finally we cover the yield forecasting pipeline.

6.1 Engineered Features for the Futures Task

Starting from the 12 raw daily CY-Bench variables per region, we create 14 additional features organised into five categories. These are driven by both patterns observed in the previous section as well as the agronomic literature.

Firstly, we follow CY-Bench's own feature design [37] by creating **threshold-based stress indicators**. We count cold days (tmin < 0°C), hot days (tmax > 35°C), and dry days (precipitation < 1 mm) over rolling 30-day windows. Indeed, hot days can be defined as > 35°C given that Tack et al. [48] show that above 34°C (in spring) wheat experiences the largest yield reduction. We further introduce "grain-fill" heat stress features (this is the stage at which the plants fill their grains from the nutrients they absorb) for the May-July window using a 30°C threshold, counting both the number of exceedance days and the cumulative excess degrees above 30°C over 30-day windows. This lower threshold is motivated by Schlenker and Roberts [46], who found that corn and soybean yield is negatively impacted beyond temperatures of 29 and 30°C, respectively. Although their study focuses on different crops, we lower the threshold to 30°C because Porter and Gawith [38] find that wheat is more sensitive to high temperatures during its reproductive phase which includes grain-fill. Finally, we also compute spring heat shock indicators (tmax > 25°C during March-May) over 30-day and 60-day windows to capture early-season stress.

Secondly, fPAR enables us to compute **vegetation anomalies** relative to an expanding day-of-year climatology: $\text{anomaly}_{d,y} = \text{fPAR}_{d,y} - \bar{\text{fPAR}}_{d,<y}$, where $\bar{\text{fPAR}}_{d,<y}$ is the historical mean for that calendar day using only prior years. The expanding window avoids information leakage from future years. We retain both a raw daily anomaly and a 30-day rolling mean. This approach follows the near-real-time anomaly detection methodology of Meroni et al. [33].

Thirdly, we compute a **drought feature** through a Standardised Precipitation Index (SPI)-like measure by z-scoring rolling 30-day and 60-day cumulative precipitation against its expanding historical distribution, following McKee et al. [32]. Positive values indicate wetter-than-normal conditions and negative values indicate drought.

Finally, a key additional feature to avoid information loss is a **weekend weather accumulator**. Indeed, since futures markets do not trade on weekends but weather events still occur, we accumulate 3-day precipitation (sum), maximum temperature, and minimum temperature on Mondays to capture the weekend weather surprise. This is motivated by the timescale mismatch observed in Section 5: agricultural data does not update over weekends while Monday returns must incorporate two days of unpriced information.

After our feature engineering, each region has 26 features (12 raw + 14 engineered). We intentionally exclude all price-derived features from our models since we want to focus on whether agricultural features from CY-Bench carry predictive power. Adding price features may lead our models to rely more on market data to make predictions as price derived features are directly impacted by different external factors and geopolitical events, which is not the case for agricultural data.

6.2 Target Construction and Temporal Alignment

As briefly mentioned in Section 4, our target is not the raw daily return but the 5-trading-day forward return:

$$y_t = \frac{P_{t+5}}{P_t} - 1 \tag{1}$$

where P_t is the roll-adjusted closing price on day t. Since the ratio roll-adjustment preserves percentage returns across contract rolls, y_t equals the compounded return a roll-following investor would experience over that window regardless of whether a roll occurred. We choose a 5-day horizon rather than a daily one because the EDA in Section 5 showed that agricultural features evolve on weekly-to-monthly timescales while daily returns may be dominated by order-flow dynamics and other noise. Indeed, Deuskar and Johnson [11] found that flow-driven returns account for 40 to 70% of market variance. This choice is also similar to Thaker et al. [49], who use a 20-trading-day horizon for their CNN-based wheat futures model. We include additional details on target construction in Appendix C.

A central challenge is that the agricultural data are available at regional (sub-national) level on a daily or sub-daily calendar, while futures prices exist only on trading days and represent a single EU-wide market. Thus, as a next step, we merge the regional agricultural features with the futures price data, by using a backward temporal join i.e. for each trading day, each region is matched to its most recent agricultural observation on or before that date. Since both datasets are fully available for the February 2003 to December 2023 period defined in Section 4, this merge introduces no missing values.

All features are standardised to zero mean and unit variance before model fitting.

Since we have data for 507 regions, we choose to fit one model per region and then aggregate the predictions to EU-level. This aggregation follows Paudel et al. [36], who showed that regional models outperform nationally aggregated ones (in their yield task). This entails that once a separate model is fitted to each region, we aggregate the 507 per-region predictions into a single EU-level forecast by using production-weighted averaging:

$$\hat{y}_{t+5}^{\mathrm{EU}} = \frac{\sum_{r=1}^{R} w_{r,t}\, \hat{y}_{t+5}^{(r)}}{\sum_{r=1}^{R} w_{r,t}} \tag{2}$$

where $w_{r,t} = \mathrm{yield}_{r,t-1} \times \mathrm{area}_{r,t-1}$ uses the previous year's production to avoid look-ahead bias. We note that for Germany, where CY-Bench provides harvest area only at census intervals, we forward-fill area within each region, which is the norm in operational European crop monitoring systems [7]. Finally, we note that by employing a one model per region scheme, we expose ourselves to the quality of the production data, which may introduce bias to our predictions given its annual frequency.

6.3 Crop Yield Forecasting Pipeline

While the primary aim of this study is to test whether CY-Bench features predict futures returns, we also implement the task these features were designed for: sub-national wheat yield forecasting. We conduct this dual-task in view of evaluating how informative the features are in a purely agricultural task. This enables us to verify that the data contains useful information for yield prediction while testing if it also contains useful information for futures price prediction.

Our yield pipeline aggregates daily CY-Bench time series into monthly summary features within each region's growing season, the are taken from the CY-Bench static crop calendars which have start-of-season (SOS) and end-of-season (EOS) dates for each region. For each region-year pair, we compute monthly means for all meteorological, soil moisture, and vegetation variables over the available months, so that each observation is a single row with columns such as the mean ndvi in october and mean fpar in december. This preserves the temporal structure of the growing season while reducing dimensionality. Following CY-Bench's lead-time framework [37], we build three separate datasets by truncating the available data at different points before harvest: end-of-season (all growing-season data available), mid-season (data truncated at $(SOS + EOS)/2$), and quarter-season (data truncated at $SOS + (EOS - SOS)/4$). These represent real-world forecasting scenarios where earlier predictions are more valuable but less accurate. We note that our monthly calendar-based aggregation differs from Paudel et al. [36] who aggregate by crop development phases using the a crop simulation model. Our approach is simpler but we believe it may provide additional granularity compared to the CY-Bench methodology given that we data for each month.

The target variable is the official reported wheat yield in tonnes per hectare. We adopt a temporal split appropriate for the 2003-2020 data window: training on 2003-2012, validation on 2013-2015 (for hyperparameter tuning only), and test on 2016-2020. The test period includes the 2018 European drought, providing a challenging evaluation. Following the cross-country differences in yield level and variability observed in Section 5, we tune hyperparameters separately for each country rather than using a single EU-wide configuration. We report normalised RMSE and R^2 at both per-region and production-weighted national levels, following CY-Bench [37].

7 Model Fitting and Tuning Methodology

In this section, we cover the methodology used to fit and tune all models for our two prediction tasks. Firstly, all models in this study are evaluated using walk-forward validation with an expanding training window. Secondly, all hyperparameters are tuned via random grid search given the high dimensionality. We cover these methodological choices prior to the individual model results as these choices underpin every model we implemented.

Walk-forward validation is a temporal technique analogous to cross-validation designed for timeseries. Rather than randomly partitioning observations into training and test folds, which would allow the model to train on future data and predict past data. This walk-forward scheme respects the chronological ordering of the timeseries dataset. For the futures price prediction task, we divide the data into three chronological partitions: training from 2003 to 2016, validation from 2017 to 2019, and test from 2020 to 2023, giving an approximate 67/14/19 split. The training cutoff is placed after 2016 so the model learns from both the 2007-2008 global food price crisis, during which wheat prices rose by 136%, and the secondary price spike in 2010-2011, giving it exposure to the most volatile periods in recent agricultural commodity history. The validation window from 2017 to 2019 covers a relatively stable period that is suitable for hyperparameter tuning without contaminating the test evaluation. The test period from 2020 to 2023 then provides a genuinely challenging out-of-sample assessment, encompassing both COVID-19 supply chain disruptions and the 2022 Russia-Ukraine conflict during which wheat prices surged to their highest levels since 2008. This split is broadly consistent with the 60/20/20 ratio used in the SustainBench crop yield benchmark and with Paudel et al. [36], who trained their ML baseline on data up to 2011-2012 and tested on subsequent years. At evaluation time, the expanding window means that for each test year

$T \in \{2020, \ldots, 2023\}$, the model is retrained using all available daily observations prior to T. This ensures that the training set grows with each subsequent test year, incorporating an increasing history of market and climate conditions while never exposing the model to future information. Each of the 507 regions is fitted independently with features standardised using a Standard Scaler fitted on the training data only, and region-level predictions are aggregated to a single daily signal via production-weighted averaging, where each region's weight reflects its lagged wheat production share.

For the yield prediction task, the temporal split is shifted earlier because subnational yield statistics end in 2020 for France and Poland. The partitions are training from 2003 to 2012, validation from 2013 to 2015, and test from 2016 to 2020, giving an approximate 56/17/28 split. The same walk-forward expanding window logic applies: for each test year $T \in \{2016, \ldots, 2020\}$, all prior region-years are used for training and the model predicts regional yield for year T. The test window includes the 2018 European drought, one of the worst on record for wheat with German yields dropping sharply, and the 2020 season which was affected by unusual spring weather patterns across Western Europe. The yield models are tuned separately per country rather than across the entire dataset, because the three countries differ substantially in the number of regions, with Germany at 251, France at 90, and Poland at 16. Predictions are aggregated to national level using production weights as described in Section 6 which follows the methodology of Paudel et al. [36].

For hyperparameter tuning, we use random search rather than grid search following Bergstra and Bengio [2], who showed that random search is more efficient than grid search in high-dimensional hyperparameter spaces because it avoids wasting evaluations on unimportant parameter combinations. Each model draws 20 random configurations from the specified parameter distributions and evaluates them over the validation period. For the futures task, the scoring criterion is directional accuracy rather than R^2, since the practical objective is to correctly predict the sign of the 5-day return rather than its magnitude. An R^2-based selection criterion would favour models that minimise squared error in absolute terms, but a model that consistently predicts the correct direction of price movement with imprecise magnitude is more useful than one that occasionally produces a precise forecast but frequently misses the direction. For the yield task, the scoring criterion is region-level Normalised Root Mean Squared Error (NRMSE), which measures the percentage prediction error relative to the mean observed yield and enables comparison across countries with different absolute yield levels.

8 Ridge and Lasso Regression

Ridge and Lasso regression are evaluated together as the linear baseline methods for both our primary prediction task of wheat futures price, as well as the secondary task of wheat yield prediction. Both are appropriate first choices for this high-dimensional tabular setting for similar but distinct reasons. Comparing them directly tests whether dense coefficient shrinkage or sparse feature selection better characterises the underlying signal in our agricultural feature set.

For the Futures Price Prediction Task, Ridge is chosen because its L2 penalty shrinks correlated coefficients toward each other rather than eliminating them, preserving the joint contribution of related agricultural indicators such as normalized difference vegetation index (NDVI), a satellite-based remote sensing method, and soil moisture that together capture crop stress conditions. Lasso complements this with an L1 penalty that drives irrelevant coefficients to exactly zero, functioning as a feature selector completely eliminating these features. If the return signal across 76 agricultural features is concentrated in a small subset, we would expect Lasso to identify and retain only those features, which should outperform Ridge by removing noise from uninformative predictors. Both models are computationally trivial to fit across all 515 regions, making them practical to run independently for every region across every walk-forward window. Both also serve as a baseline method to compare against, where a strong result from either model would imply the futures return signal is largely linear and that escalating to ensemble methods potentially adds no value, while a poor result from both motivates the tree-based architectures that follow. Finally, regularised regression methods feature prominently in the literature we reviewed [36, 52], making this a natural point of methodological comparison between our study and established benchmarks.

The sole hyperparameter for Ridge Regression is the regularisation strength α, which controls the trade-off between fit and coefficient shrinkage. We sample α from a log-uniform distribution [2] over a range of [0.01,100]. The log-uniform distribution is appropriate here because the effect of regularisation on model performance is multiplicative rather than additive, meaning that doubling α from 0.1 to 0.2 has a similar effect to doubling it from 10 to 20, ensuring that we spend equal time looking at different orders of magnitude and not focusing too much on higher α while neglecting the lower bound. The lower bound of 0.01 corresponds to Ordinary Least Squares behaviour, where regularisation is minimal, and the upper bound of 100 enforces strong shrinkage appropriate for a noisy financial target. Twenty random draws are evaluated over the validation period from 2017 to 2019 [2], with directional accuracy used as the selection criterion [30] rather than R^2, since the practical objective is to correctly predict the sign of the 5-day return rather than its magnitude. This is an important distinction since by using the directional accuracy we can more accurately predict if the model is predicting trends and patterns rather than trying to predict precise market movements that could result in poor R^2 results, regardless if the direction was correct.

Lasso uses a narrower log-uniform range of [1e-5, 0.1], reflecting that L1 regularisation operates at a fundamentally different scale compared to Ridge. Lasso coefficients reach zero at much smaller α values compared to Ridge coefficients, so applying the same upper bound of 100 would produce a degenerate all-zero model. Lasso, like Ridge, uses 20 random draws over the validation period from 2017 to 2019 with directional accuracy as the selection criterion for the same reasons as Ridge as stated above.

The optimal Ridge configuration is α = 63.51, situated in the upper quarter of its log-uniform search range. This reflects that strong regularisation is beneficial when the target is noisy and many features carry weak signal. Lasso's optimal α value is 0.000146, very close to the lower bound of its range, resulting in a strikingly different result. This indicates that even very mild L1 shrinkage causes the model to deteriorate due to Lasso eliminating certain features that are helpful and hold predictive importance. We can interpret from this that signal features are not concentrated in a sparse feature subset. Therefore, a more dense solution is preferred, which is consistent with Ridge's strong regularisation requirement. Ridge achieves a validation directional accuracy of 49.93% and R^2 of -0.047. Lasso achieves a marginally better validation directional accuracy of 50.72% and R^2 of -0.042. Neither model shows meaningful predictive promise for wheat futures price movement based on these validation results.

Both models are fit using a walk-forward expanding window scheme, trained independently on each of the 515 regions using all daily observations prior to each test year, with a Standard Scaler fitted on training data only and predictions aggregated via production-weighted averaging [36].

On the test set from 2020 through 2023, Ridge achieves R^2 = -0.0212, directional accuracy of 46.97%, and RMSE of 0.0409. Lasso performs marginally better across all metrics with R^2 = -0.0183, directional accuracy of 47.55%, and RMSE of 0.0409. Both models see directional accuracy fall below their validation results and RMSE approximately double from 0.020 to 0.041, signs that the performance on the test set is reduced compared to the validation results. The RMSE increase reflects the substantially larger absolute return magnitudes during the test period, which encompasses COVID-19 supply chain disruptions and the 2022 Russia-Ukraine conflict during which wheat prices surged to their highest levels since 2008, rather than purely model degradation. The negative R^2 for both models confirms that neither outperforms the unconditional mean, showing that both Ridge and Lasso Regression are no better than just predicting the mean return every day for the wheat futures price prediction task, consistent with what would be expected under semi-strong market efficiency [15]. The marginal Lasso advantage over Ridge on the futures task suggests at most a very weak degree of feature sparsity in the predictive signal, but the difference is too small to be practically meaningful given the overall poor results from both models.

For the secondary task of Yield Prediction, both models are tuned separately per country over the validation period from 2013 to 2015 with region NRMSE as the scoring criterion. Ridge optimal α is 75.79 for Germany, 21.37 for France, and 75.79 for Poland. Germany's high value reflects its 251 different regions being widest relative to training observations, requiring stronger shrinkage to prevent overfitting. France's lower value reflects its 90 regions providing a better estimation problem where more feature variation can be retained usefully. For Lasso, optimal alphas are 0.0126 for Germany, 0.00315 for France, and 0.0847 for Poland. All α values are small relative to the Ridge values, again consistent with Lasso needing much less regularisation than Ridge to achieve its best performance. On the validation set, Lasso slightly outperforms Ridge for Germany, 12.11% vs 13.03% NRMSE scores, Ridge narrowly wins for France with 10.67% vs 10.88%, and Ridge wins more clearly for Poland at 16.65% vs 17.44%.

Both models are then evaluated on the test period from 2016 through 2020 in a walk-forward scheme, training on all prior region-years and predicting regional yield with predictions aggregated to national level using production weights.

These final test set yield results show that for the end-of-season, we can see performance varies considerably by country. For Germany, Ridge is the worst-performing linear model with region NRMSE at 18.76% and EU NRMSE at 13.39% while Lasso improves substantially getting a region NRMSE of 16.32% and EU NRMSE score of 9.14%, confirming that L1 sparsity is beneficial when the end-of-season feature set spans 351 columns and many are redundant. For France, both models perform similarly at regional level, with Ridge 21.77% and Lasso 22.25%, with both falling well short of Paudel et al. [36]'s best results of approximately 6% for national NRMSE, which we can attribute to their use of WOFOST (water-limited yield biomass, leaf area index, development stage), compared to ours where we use only raw CY-Bench climate and remote sensing variables aggregated by calendar month. Poland, which was not included in Paudel et al.[36], shows that Ridge achieves the best end-of-season result of any model in any country with region NRMSE at 12.61% and EU NRMSE at 4.08%, while Lasso degrades to 17.30% regionally and 13.47% at EU level. This reversal from the validation ranking suggests that for Poland's small 16-region sample, Ridge's dense shrinkage is more stable out-of-sample than Lasso's variable selection, which is sensitive to which features are retained across different training windows. The Polish Ridge result should be treated cautiously given the small sample size and inability to compare against the Paudel et al.[36] results. The year-by-year national aggregate predictions for the best model per country are shown in Figure E10.

Across lead times, the pattern for Germany is consistent for both models. Ridge achieves its best German performance at mid-season (17.13%) rather than end-of-season (18.76%), with quarter-season only marginally worse at 17.24%. Lasso follows the same pattern, peaking at mid-season (15.65%) relative to end-of-season (16.32%) and quarter-season (16.40%). This suggests that for Germany, the mid-season feature set strikes the best balance between containing enough growing-season information and avoiding the noise introduced by late-season variables. France degrades differently, with Ridge moving from 21.77% at end-of-season to 25.46% at mid-season and then back to 21.78% at quarter-season. This suggests that mid-season features for France are particularly noisy, a sharp contrast to that of Germany's results. Poland shows sharp decline at mid-season for both models with Ridge going from 12.61% at end-of-season to 18.90% at mid-season. Lasso goes from 17.30% at end-of-season to 18.40% at mid-season, reinforcing the instability of estimates with only 16 regions. Across all lead times and countries, the degradation from end-of-season toward earlier lead times is at most modest for Germany and France, as seen in Figure D9. These results are consistent with the finding of Paudel et al. [36] that structural yield drivers such as soil characteristics and long-term climate patterns carry predictive power throughout the season rather than only at harvest.

Overall, Lasso and Ridge serve as modest and acceptable baselines to compare against. Lasso outperforms Ridge for Germany across all lead times while Ridge is more stable for Poland, and the two are comparable for France. This pattern, the results we achieved as discussed in this section, and the inability of both linear models to capture non-linear interactions between weather, soil, and vegetation features, motivates the examination of tree-based ensemble methods next, which can represent relationships that neither Ridge nor Lasso are capable of modelling.

9 Random Forest and Extra Trees Regressors

Random Forest and ExtraTrees are evaluated together as the tree-based ensemble methods in our pipeline, following directly from the linear baseline established by Ridge and Lasso. Both methods are motivated by the same core limitation identified in the previous sections finding that the linear models cannot represent interactions between features that together signal crop stress in ways that no single feature captures alone. Tree-based ensembles address this by partitioning the feature space through successive splits, naturally capturing these non-linear interactions without requiring them to be specified manually, and have been shown to remain competitive with more complex deep learning architectures on tabular datasets [20].

For the Futures Price Prediction Task, Random Forest (RF) is chosen for three reasons. With its bagging approach for training each tree on a bootstrapped sample of observations and averaging predictions across trees, we can aim to reduce variance relative to a single tree [3], which is particularly valuable in a noisy financial return prediction task where individual trees would overfit heavily to the training window. The random feature subsampling at each split de-correlates the trees [3], preventing any single dominant agricultural feature from driving all predictions and encouraging the ensemble to exploit the full breadth of the 76 available features. From a practical perspective, RF's robustness to hyperparameter choice makes it a reliable step up from linear models without requiring extensive tuning to obtain useful results [40]. Aditionally, RF appears prominently in the yield prediction literature we reviewed [36, 37, 52], making it a natural methodological bridge between the crop yield benchmarks and our futures application.

ExtraTrees is evaluated alongside RF as a direct comparison. The key distinction is that ExtraTrees uses random rather than optimal split thresholds at each node, introducing additional randomisation that further reduces variance at the cost of some increase in bias [18]. In a setting with many weakly informative features and a noisy target, this extra randomisation may be beneficial by preventing the ensemble from overcommitting to any particular feature split that happens to perform well in training but does not generalise during testing.

The hyperparameter search for both models covers n_estimators from [200,500] for RF and fixed at 500 for ExtraTrees. max_depth is set to set of 3,5, or 8, min_samples_leaf at 10, 20, or 40, with max_features set to square root, log base 2, 0.3, or 0.5. All values are discrete rather than continuous because ensemble behaviour is not sensitive to small perturbations in these parameters [40]. The meaningful distinctions are between shallow and deep trees, few and many estimators, and narrow versus wide feature subsets. The max_depth upper bound of 8 prevents deep trees that would memorise individual training windows, which is critical in a walk-forward scheme where the training set grows by only one year at each step. The min_samples_leaf lower bound of 10 enforces a minimum leaf size that prevents extremely specific splits on small subsets of the 76 agricultural features. Both models use 20 random draws over the validation period from 2017 through 2019 with directional accuracy as the selection criterion for the same reasons we used directional accuracy as stated in the Ridge and Lasso Regression section.

The optimal RF configuration is n_estimators=500, max_depth=5, min_samples_leaf=40, max_features=0.5, achieving a validation directional accuracy of 49.28%. The large min_samples_leaf of 40 and shallow max_depth of 5 together indicate that strong regularisation is needed even for tree ensembles on this task [27, 40]. This shows the model benefits from broad, large splits rather than finer grained partitions, consistent with the weak nature of the agricultural signal. For ExtraTrees, the optimal configuration is n_estimators=500, max_depth=5, min_samples_leaf=20, max_features=sqrt, achieving a validation directional accuracy of 46.80%. The lower min_samples_leaf relative to RF is consistent with ExtraTrees additional randomisation at split points providing implicit regularisation [18], reducing the need to enforce large leaf sizes explicitly.

Both models are fit using the same walk-forward expanding window scheme as the linear models, trained independently across all 515 regions with a Standard Scaler on training data only and predictions aggregated via production-weighted averaging.

On the test set from 2020 through 2023, RF achieves R^2 = -0.0099, directional accuracy of 48.43%, and RMSE of 0.0407. ExtraTrees achieves R^2 = -0.0074, directional accuracy of 50.98%, and RMSE of 0.0407. Both models improve over Ridge and Lasso on every metric. Notably, ExtraTrees is the only model in our entire pipeline to exceed 50% directional accuracy on the test set, suggesting that the additional randomisation in ExtraTrees split selection provides a marginal but real advantage in this noisy prediction environment. However, both R^2 values remain negative, confirming that neither ensemble model outperforms the unconditional mean on the futures task, similarly to Ridge and Lasso. The RMSE values are the same when rounded to two decimal places, and nearly the same to those of the linear models, again reflecting the dominant effect of the high-volatility 2022 period rather than model-specific behaviour.

For the Yield Task, RF and ExtraTrees are tuned separately, per country, over the validation period from 2013 to 2015 with region NRMSE as the scoring criterion, using the same hyperparameter grid as for the futures task but with max_depth extended to include 12 and None (unlimited) to allow the trees to grow deeper on the tabular annual yield dataset, which has more structured signal than daily futures returns.

The optimal configurations differs meaningfully across countries. For Germany, RF selects n_estimators=500, max_depth=8, min_samples_leaf=20, max_features=0.5, achieving a validation NRMSE of 11.36%. ExtraTrees selects n_estimators=500, max_depth=12, min_samples_leaf=5, max_features=0.5 with validation NRMSE of 11.76%. Both models choose deeper trees and smaller leaf sizes compared to the futures task, reflecting the more structured annual yield signal where more precise feature interactions improve prediction. For France and Poland, both models independently select max_depth=None and min_samples_leaf=2, allowing unrestricted tree growth. This may reflect that France's 90 regions and Poland's 16 regions provide smaller training sets where deeper trees do not overfit as severely, or that the yield signal in these countries is concentrated in strong feature interactions that require deeper splits to capture. RF acheives a regional NRMSE of 12.25% and ExtraTrees is at 12.66% on France. For Poland, RF is 17.43% and ExtraTrees is 17.40%. The higher validation NRMSE for France and Poland compared to Germany, combined with the need for unrestricted tree depth, suggests that the ensemble models are less effective in settings with fewer regions and less structured training data.

Both models are evaluated on the test period of 2016 through 2020 in a walk-forward scheme, training on all prior region-years with predictions aggregated to national level using production weights.

At end-of-season, both tree ensembles substantially outperform the linear models for Germany. RF achieves a region NRMSE of 13.57% and EU NRMSE of 4.57%, while ExtraTrees achieves 13.00% and 4.24% respectively, making ExtraTrees the best-performing model for Germany across all methods. These results are broadly comparable to Paudel et al. [36]'s German baseline of approximately 15% regional NRMSE and confirm that non-linear interactions between weather, soil, and vegetation features are important for German yield prediction. For France, RF achieves region NRMSE of 17.97% and EU NRMSE of 14.34%, while ExtraTrees achieves 17.77% and 13.90%, again with ExtraTrees narrowly ahead. Both remain below the Paudel et al. [36] benchmark of approximately 6% national NRMSE for the same WOFOST feature quality reasons discussed in the previous section. For Poland, RF achieves region NRMSE of 16.90% and EU NRMSE of 12.34%, while ExtraTrees achieves 16.24% and 12.12%. Both are substantially worse than Ridge's 12.61% regional and 4.08% EU result, confirming that Ridge's Poland performance was driven by the small sample size rather than a genuine linear signal advantage.

Across lead times, both tree models show the same gradual degradation from end-of-season toward earlier lead times that we observed for the linear models, consistent with Paudel et al. [36]'s finding that structural yield drivers dominate throughout the season. For Germany, ExtraTrees moves from 13.00% at end-of-season to 14.60% at mid-season and 15.53% at quarter-season, a degradation of only 2.5 percentage points across all three lead times. RF follows an almost identical pattern at 13.57%, 14.56%, and 15.66%. For France, ExtraTrees moves from 17.77% to 19.12% to 20.23%, and RF from 17.97% to 20.29% to 21.26%. Again, a gradual and consistent increase. Poland shows sharper degradation for both tree models, similar to the pattern seen with linear models and reinforcing the small-sample instability issue. Across all countries and lead times, ExtraTrees outperforms or matches RF, confirming that additional split randomisation is consistently beneficial for this yield prediction task, as illustrated in Figure D9. The year-by-year national aggregate predictions are shown in Figure E10.

Overall, the tree-based ensembles establish a substantially improved baseline over the linear models for yield prediction, while delivering only marginal improvements on the futures task. The gap between the two tasks reinforces the core finding that agricultural features carry predictive information about crop yields but that information does not translate into exploitable futures price signal, consistent with semi-strong market efficiency [15]. The neural network is examined next to assess whether greater model expressiveness can close this gap further.

10 Neural Networks

The neural network is the final model in our pipeline and serves a specific diagnostic role: it acts as a capacity upper bound on the class of learnable functions from agricultural features to futures returns. If a sufficiently flexible nonlinear model with universal approximation capability [17] cannot extract predictive signal from CY-Bench features, the failure is unlikely to be attributable to insufficient model expressiveness in the preceding tree-based experiments. Three considerations motivate its inclusion. First, deep networks can represent arbitrary compositions of input interactions without the axis-aligned partitioning constraint that limits individual trees. If the relationship between agricultural variables and returns involves smooth, high-order nonlinearities, for instance a drought signal that only manifests when temperature, precipitation, and NDVI jointly cross thresholds, a neural network is better equipped to detect it than an additive ensemble of shallow trees. Second, Grinsztajn, Oyallon, and Varoquaux [20] showed at NeurIPS 2022 that tree-based methods remain competitive with deep learning on medium-scale tabular data, making this an empirical question rather than a settled one for our specific dataset. Third, from a market efficiency perspective, confirming that even a high-capacity learner produces negative out-of-sample R^2 strengthens the conclusion that the failure to predict returns reflects information already incorporated into prices rather than a limitation of the model class.

We evaluate four feedforward configurations spanning the capacity spectrum from heavily constrained to moderately deep, all using ReLU activations and a single linear output neuron for regression. NN-Small uses a single hidden layer of 32 units with dropout 0.3 and weight decay of 0.01. NN-Medium uses two hidden layers of 64 and 32 units with the same dropout and weight decay. NN-Deep uses three hidden layers of 128, 64, and 32 units with dropout raised to 0.4 and learning rate reduced to 5×10^{-4} to stabilise gradient flow through the deeper architecture. NN-Heavy-Reg uses the same 64-32 architecture as NN-Medium but with dropout increased to 0.5 and weight decay raised to 0.05, providing the strongest regularisation. The architectural choices are deliberately conservative, restricting depth to at most three hidden layers with at most 128 units, because each region provides only approximately 3,500 daily training observations and the effective dimensionality of the agricultural signal is low. Dropout rates range from 0.3, the default recommended by Srivastava et al. [45], to 0.5, where Gal and Ghahramani [19] interpret the thinned ensemble as approximate variational Bayesian inference. Weight decay imposes Ridge-like shrinkage on the parameters. All configurations share early stopping with patience of 15 epochs on a temporal validation set consisting of the last 20% of each training window, a maximum of 150 training epochs, and a batch size of 64 where small batches inject gradient noise shown to aid generalisation [53]. The optimiser is Adam [25] with weight decay, chosen over SGD because convergence speed matters when training 515 region-specific models per test year.

The best-performing configuration is NN-Medium (64-32), which achieves a validation directional accuracy of 51.27%. The two-layer architecture with moderate dropout strikes the best balance between capacity and regularisation. Adding a third layer in NN-Deep does not improve performance, consistent with the observation that increasing depth beyond the point where training error is already near-zero introduces optimisation difficulty without reducing test error. The heavily regularised variant NN-Heavy-Reg performs marginally worse than the standard NN-Medium, suggesting that dropout of 0.5 combined with weight decay of 0.05 over-constrains the already limited capacity.

The model is fit using the same walk-forward expanding window scheme as the linear and tree-based models, trained independently for each of the 515 regions using all daily observations prior to each test year, with a Standard Scaler fitted on training data only and predictions aggregated via production-weighted averaging.

On the test set from 2020 through 2023, NN-Small achieves $R^2 = -0.0073$, directional accuracy of 51.37%, and RMSE of 0.040648. NN-Medium achieves $R^2 = -0.0061$, directional accuracy of 51.27%, and RMSE of 0.040624. NN-Deep achieves $R^2 = -0.0074$, directional accuracy of 51.27%, and RMSE of 0.040650. NN-Heavy-Reg achieves $R^2 = -0.0064$, directional accuracy of 51.27%, and RMSE of 0.040629. All four configurations marginally outperform the tuned ExtraTrees baseline of $R^2 = -0.0074$ and RMSE = 0.0407, though the differences are economically negligible, corresponding to a prediction improvement of approximately 0.06% of the unconditional return standard deviation. Notably, all four neural network configurations and the ExtraTrees model cluster around 51% directional accuracy, barely above the 50% naive baseline, indicating no economically meaningful directional signal. Every model in the entire pipeline, linear, tree-based, and neural, produces negative R^2, confirming that CY-Bench agricultural features do not contain exploitable signal for daily futures return prediction. This is consistent with semi-strong market efficiency [15]: publicly available satellite and meteorological data are already reflected in wheat futures prices by the time they appear in the CY-Bench dataset. The fact that increasing model capacity from a single 32-unit layer to a 128-64-32 deep network does not improve predictive performance provides strong evidence that the bottleneck is signal quality, not model expressiveness, following the diagnostic framework of Goodfellow et al. [17].

Now looking at the yield prediction task, we significantly escalate model capacity by adopting a pooled approach where a single neural network is trained across all 357 regions simultaneously, rather than fitting one model per region as in the previous methods. To prevent the curse of dimensionality associated with one-hot encoding hundreds of regional identifiers, we employ entity embeddings [16] that learn continuous dense representations of spatial heterogeneity for both region and country variables, with embedding dimensions of 16 and 4 respectively. We evaluate three architectures. The

Pooled MLP uses two hidden layers of 128 and 64 units with LayerNorm, ReLU activations, and dropout of 0.3, where numerical features are concatenated with the regional and country embeddings. The FT-Transformer [21] projects each scalar agricultural feature into a 16-dimensional token and applies a single Transformer encoder layer with 4 attention heads to model inter-feature dependencies, aggregating via a CLS token. The 1D-CNN, inspired by the CNN-RNN framework of Khaki, Wang, and Archontoulis [26], applies two convolutional layers with kernel sizes of 5 and 3 and 64 channels with BatchNorm, treating the feature vector as a 1D sequence before pooling and concatenating with entity embeddings. All three models use AdamW [28] with decoupled weight decay rather than standard Adam, because Loshchilov and Hutter showed that L_2 regularisation interacts poorly with adaptive learning rate methods unless the weight decay is applied directly. The learning rate is reduced by a factor of 0.5 when validation loss plateaus for 10 consecutive epochs, with early stopping at patience of 30 epochs.

All models are evaluated on the test period from 2016 through 2020 in a walk-forward scheme, training on all prior region-years with the final 20% reserved as temporal validation for early stopping. Inputs are Standard Scaler normalised and region and country identifiers are integer-encoded for the embedding layers.

At end-of-season, the 1D-CNN achieves the strongest per-region performance with region $R^2 = +0.410$ and region RMSE of 1.140 t/ha, indicating that the convolutional filters successfully extract localised patterns across the monthly aggregated features. The Pooled MLP performs comparably at region $R^2 = +0.371$ and RMSE of 1.177 t/ha, confirming that entity embeddings alone provide substantial cross-regional information sharing. The FT-Transformer underperforms both alternatives with a negative region R^2 of -0.116 and RMSE of 1.568 t/ha. With only approximately 4,600 training observations across 357 regions and 13 years, the attention mechanism is likely over-parameterised: each of the 347 features requires learned query, key, and value projections, resulting in a parameter count disproportionate to the available data. This is consistent with the finding of Gorishniy et al. [21] that FT-Transformers require larger datasets to realise their potential. However, all three architectures produce negative EU-level R^2, with the MLP at -0.820, the 1D-CNN at -1.342, and the FT-Transformer at -5.789, meaning that the production-weighted national aggregates are worse than predicting the historical mean yield. This contrasts with the tree ensembles where ExtraTrees achieved positive EU-level R^2. The discrepancy arises because the pooled models make correlated errors across regions within the same country: when the network misestimates the effect of a country-wide weather anomaly, the error propagates to all regions sharing that country embedding and production-weighted aggregation amplifies rather than diversifies the mistake. The per-region tree models, trained independently, produce less correlated errors that partially cancel under aggregation.

At the per-country level, the results are mixed. For Germany, the Pooled MLP achieves a region NRMSE of 12.20%, marginally outperforming ExtraTrees at 13.00%, which is the only case where a neural network beats the tree baseline. Germany's 251 NUTS3 regions provide the largest sub-sample, giving the entity embeddings sufficient signal to learn meaningful spatial representations. For France, the best neural network is the 1D-CNN at region NRMSE of 21.68%, which is weaker than the ExtraTrees baseline of 17.77%, suggesting that the 90 French regions do not provide enough within-country variation for the convolutional filters to generalise. Poland presents the starkest contrast: the MLP's region NRMSE of 45.86% is nearly four times the ExtraTrees baseline of 16.24%. With only 16 NUTS2 regions, the embedding layer has far too few examples to learn distinct regional representations, and the pooled model effectively collapses Polish predictions toward the European mean. These per-country deep learning results are shown in Figure D9. The year-by-year national aggregate predictions for the best model per country are shown in Figure E10.

Overall, the neural networks deliver the best R^2 on the futures task across the entire pipeline, but all results remain negative and economically indistinguishable from the tree-based models, confirming that the CY-Bench agricultural features carry no exploitable signal for daily return prediction. For yield prediction, the pooled deep learning approach achieves competitive regional accuracy for Germany, where the large number of NUTS3 regions supports entity embedding learning, but underperforms the independently trained tree ensembles for France and Poland and at the EU aggregate level. These results confirm the finding of Grinsztajn et al. [20] that tree-based methods remain competitive with deep learning on medium-scale tabular data, particularly when the number of training observations per entity is small. The entity embedding strategy partially compensates for limited per-region data by enabling cross-regional learning, but this benefit is realised only when the country sub-sample is large enough to support the additional parameters. The model comparison section that follows consolidates results across all four methods to draw broader conclusions.

11 Model Result Comparison

Table 1 presents the full test set results across all models for the wheat futures price prediction task. The most immediate observation is that the range of performance across all models is extremely narrow. R^2 values span only from -0.0212 at the worst end on the Ridge Regression model to -0.0061 as the best result with NN-Medium(64-32). RMSE values differ by less than 0.0004, and directional accuracy ranges from 46.97% with Ridge to 51.37% with NN-Small (32). All models produce negative R^2, confirming that no method in our pipeline outperforms the unconditional mean predictor, consistent with semi-strong market efficiency [15]. Despite the overall poor predictive performance, there is a meaningful and consistent ordering across model types that warrants interpretation. Linear models perform worst, with Ridge at 46.97% directional accuracy and Lasso at 47.55%. The tree ensembles improve on this, with RF at 48.43% and ExtraTrees reaching 50.98%, the only non-neural model to exceed 50% directional accuracy. Neural network variants then cluster just above ExtraTrees, with the best result of 51.37% directional accuracy from NN-Small.

Table 1 Futures price prediction test set results (2020-2023), all models.

Model	R^2	Directional Accuracy	RMSE
Ridge	−0.0212	46.97%	0.04093
Lasso	−0.0183	47.55%	0.04087
RF	−0.0099	48.43%	0.04070
ExtraTrees	−0.0074	50.98%	0.04065
NN-Small (32)	−0.0073	**51.37%**	0.04065
NN-Medium (64-32)	**−0.0061**	51.27%	**0.04062**
NN-Deep (128-64-32)	−0.0074	51.27%	0.04065
NN-Heavy-Reg (64-32)	−0.0064	51.27%	0.04063

The improvement from linear to tree to neural models follows the well-established bias-variance tradeoff [22] where linear models are too constrained to capture even the weak non-linear interactions present in the agricultural features, while tree ensembles and neural networks have sufficient capacity to detect marginal signal. The fact that the simplest neural network, a NN-Small with a single 32-unit layer, achieves the highest directional accuracy, while the deepest network matches ExtraTrees rather than improving further, is consistent with the finding of Grinsztajn et al. [20] that on tabular datasets, additional model complexity does not reliably translate into better performance and can introduce unnecessary variance. The near identical RMSE values across all models reflect the dominant influence of the high-volatility 2022 period rather than any model-specific characteristic. The interpretation is that the agricultural features from CY-Bench contain a very weak directional signal in wheat futures returns that is only detectable by ensemble methods and neural networks but too weak for linear models while still being insufficient for any model to generate meaningful predictive accuracy.

Table 2 presents end-of-season yield prediction results across all models. For Ridge, Lasso, Random Forest, and ExtraTrees, the results show a clear and consistent ordering for Germany and France, with tree ensembles substantially outperforming linear models. ExtraTrees achieves the best regional NRMSE for both Germany (13.00%) and France (17.77%), improving on Ridge, the worst model, by 5.76 and 4.00 percentage points respectively. The region R^2 values reinforce this pattern showing Ridge produces a strongly negative R^2 of -0.4976 for Germany, reflecting a failure to explain regional variation when faced with 251 regions and a 351-column feature matrix, while ExtraTrees achieves a positive R^2 of +0.2807 for the same setting [22]. For France, both tree models achieve R^2 above +0.53, confirming that non-linear feature interactions are informative. The ordering of Ridge being the worst, Lasso intermediate, RF and ExtraTrees being the best mirrors the bias-variance hierarchy observed in the futures task. Poland only has 16 regions so it is hard to derive any substantial claims, but for the smaller region set Ridge does perform the best.

Table 2 Yield prediction test set results (2020-2023), all models.

Country	Metric	Ridge	Lasso	RF	ExtraTrees
Germany	Region NRMSE	18.76%	16.32%	13.57%	**13.00%**
	EU NRMSE	13.39%	9.14%	4.57%	**4.24%**
	Region R^2	−0.4976	−0.1339	+0.2169	+0.2807
France	Region NRMSE	21.77%	22.25%	17.97%	**17.77%**
	EU NRMSE	17.62%	18.79%	14.34%	**13.90%**
	Region R^2	+0.3130	+0.2821	+0.5314	+0.5419
Poland	Region NRMSE	**12.61%**	17.30%	16.90%	16.24%
	EU NRMSE	**4.08%**	13.47%	12.34%	12.12%
	Region R^2	+0.5711	+0.1921	+0.2290	+0.2887
		MLP-Small	**FT-Transformer**	**1D-CNN**	
All	Region R^2	+0.3711	−0.1164	+0.4098	—
	Region RMSE	1.1768	1.5679	1.1401	—
	EU R^2	−0.8202	−5.7892	−1.3415	—

Among the deep learning models, the 1D-CNN achieves the best regional R^2 of +0.4098 and lowest regional RMSE of 1.1401, outperforming MLP-Small and substantially outperforming the FT-Transformer. The strong regional performance of the 1D-CNN and MLP relative to the FT-Transformer is consistent with findings in the tabular deep learning literature that transformers require substantially larger datasets to outperform simpler architectures [20].

12 Conclusion

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13 Generative AI Statement

Appendix A Excluded Regions for the Wheat Futures Prediction Task

Table A1 Classification of Select NUTS Regions by Primary Geographic and Economic Function

Category	NUTS Code	Region Name	Rationale & Source
Urban Center	FR101	Paris	Capital city, dense urban core [23]
	FR105	Hauts-de-Seine	High-density inner Parisian suburb [23]
	DE300	Berlin	German capital and city-state [9]
Logistic Hub	DE600	Hamburg	3rd largest port in Europe [39]
	DE501	Bremen	Major maritime/freight center [9]
	DE502	Bremerhaven	Key international seaport [4]
	DE945	Wilhelmshaven	Coastal industrial hub [24]
Non-wheat Producing (Tropical / Overseas)	FRY10	Guadeloupe	Tropical climate; sugarcane/bananas [1]
	FRY20	Martinique	Tropical climate; sugarcane/bananas [1]
	FRY30	French Guiana	Equatorial rainforest climate [1]
	FRY40	La Réunion	Tropical island climate [1]
	FRY50	Mayotte	Tropical island climate [1]

Appendix B Additional EDA Figures

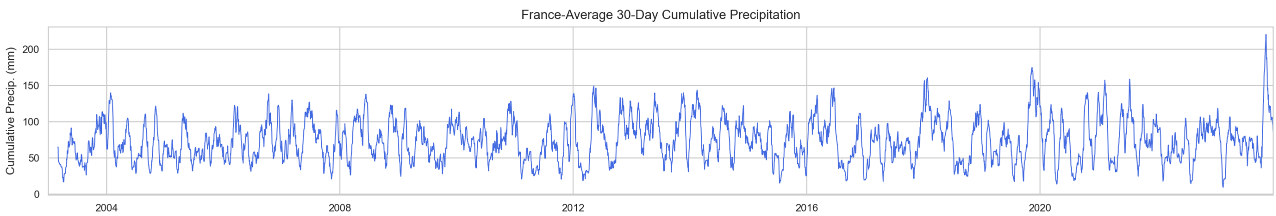

Fig. B1 France-average 30-day cumulative precipitation (2003-2023). Precipitation fluctuates on a seasonal cycle with substantial inter-annual variability but, unlike NDVI, lacks a consistent repeating pattern, reflecting the stochastic nature of rainfall events.

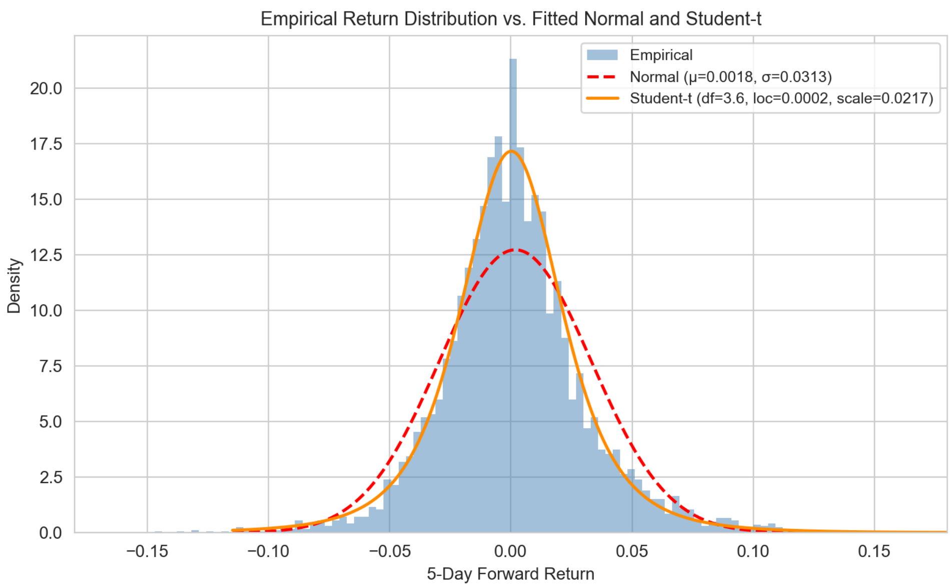

Fig. B2 Empirical 5-day return distribution compared with fitted Normal and Student-t densities. The fitted Student-t has 3.6 degrees of freedom, confirming substantially heavier tails than the Gaussian benchmark.

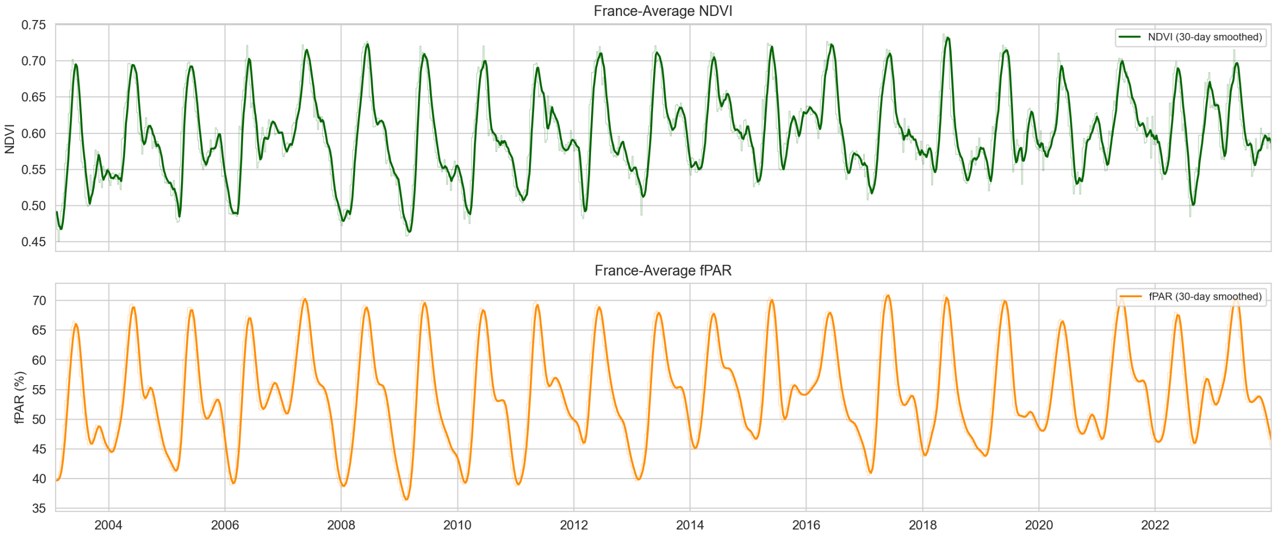

Fig. B3 France-average NDVI (top) and fPAR (bottom), 2003-2023. Both indicators track each other closely ($r = 0.93$ on 30-day smoothed series) but diverge during cloud-contaminated periods due to different gap-filling methods, justifying retaining both in the feature set.

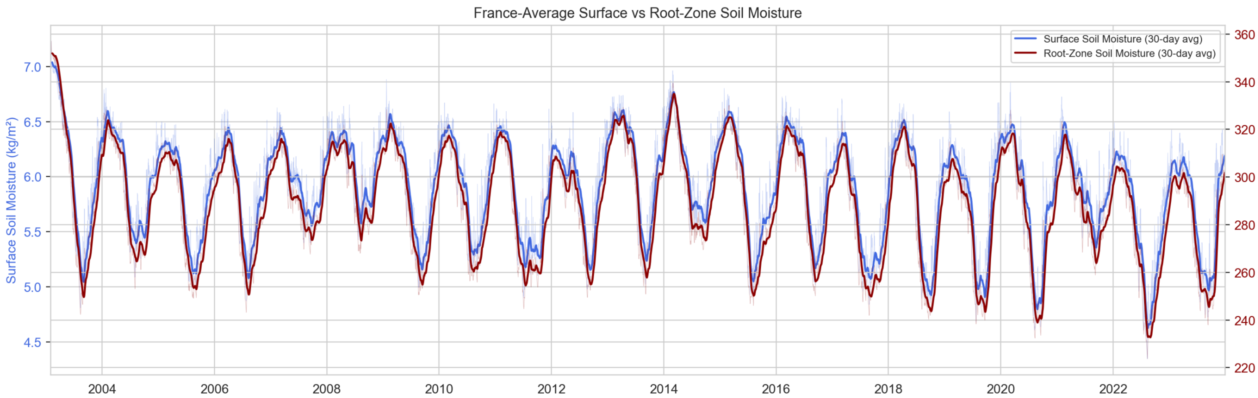

Fig. B4 France-average surface soil moisture (blue, left axis) and root-zone soil moisture (red, right axis), 2003-2023. Surface moisture responds rapidly to individual precipitation events while root-zone moisture evolves more slowly, capturing different timescales of water availability despite near-perfect correlation ($r = 0.99$) in smoothed trends.

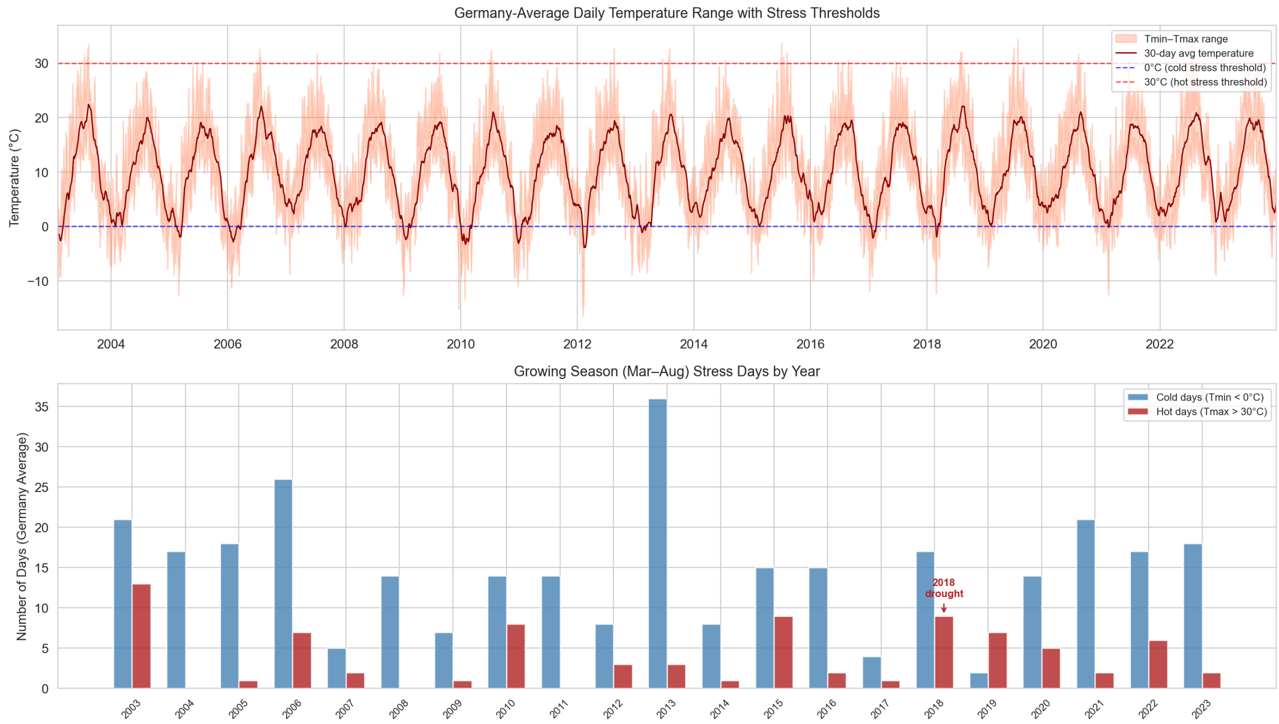

Fig. B5 Germany-average daily temperature range with stress thresholds (top) and annual growing-season (Mar-Aug) stress day counts (bottom). The 2018 European drought produced 2.5× the average number of hot stress days (Tmax > 30°C), motivating the threshold-based stress indicators in Section 6.

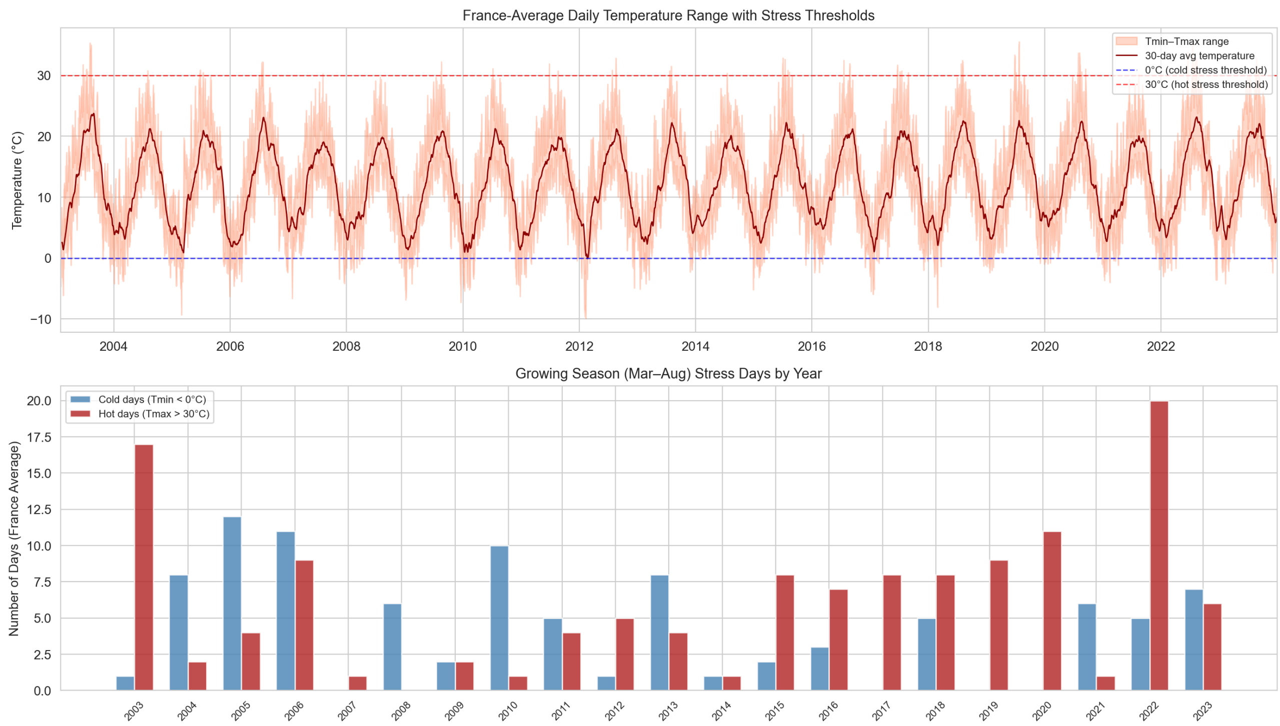

Fig. B6 France-average daily temperature range with stress thresholds (top) and annual growing-season stress day counts (bottom). France shows higher baseline hot-day frequency than Germany, with 2003 and 2022 as the most extreme heat years.

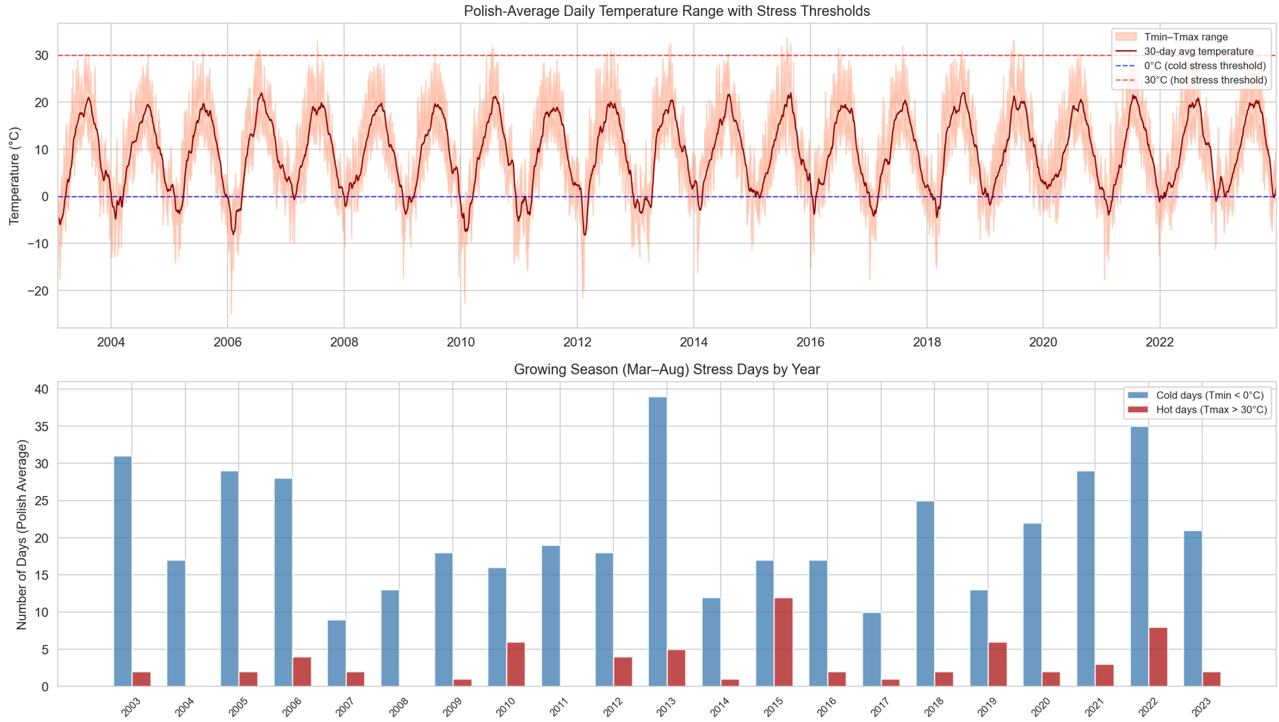

Fig. B7 Poland-average daily temperature range with stress thresholds (top) and annual growing-season stress day counts (bottom). Poland exhibits substantially more cold stress days than France or Germany, reflecting its continental climate, while hot stress events are comparatively rare.

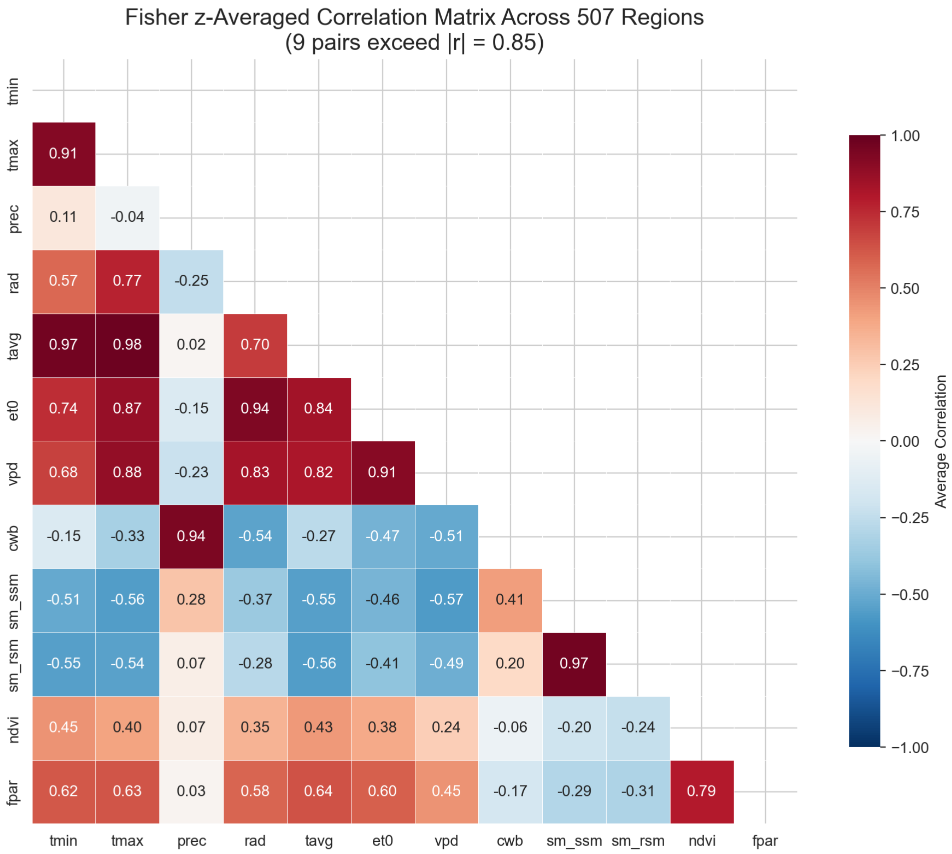

Fig. B8 Fisher z-averaged correlation matrix of the 12 raw CY-Bench variables across 507 regions. Nine feature pairs exceed $|\rho| = 0.85$, concentrated among temperature-derived variables. This motivates the correlation-based filtering applied before modelling.

Appendix C Target Construction

Apart from the timescale mismatch of daily returns and agricultural variables as well as the noise potentially present in returns at a daily scale, we also account for the fact that futures potentially have higher volatility once they approach expiration.

Indeed, the Samuelson hypothesis [44] predicts that futures price volatility increases as contracts approach expiration. Since we are looking at the closest maturing futures contract as our predictor, this implies that near-term price movements may display high variance. This further justifies our 5-day time window choice. Indeed, Duong and Kalev [12] tested this hypothesis in 20 futures markets and found strong support specifically for agricultural commodities. Daal et al. [10] also confirm that the Samuelson effect is more pronounced for commodities rather than financial futures, however, they find the effect to be absent from most contracts they test. Overall, a weekly horizon enables us to reduce this higher-variance while remaining a short enough timeframe to be operationally relevant.

Appendix D NRMSE Results for yield forecasting

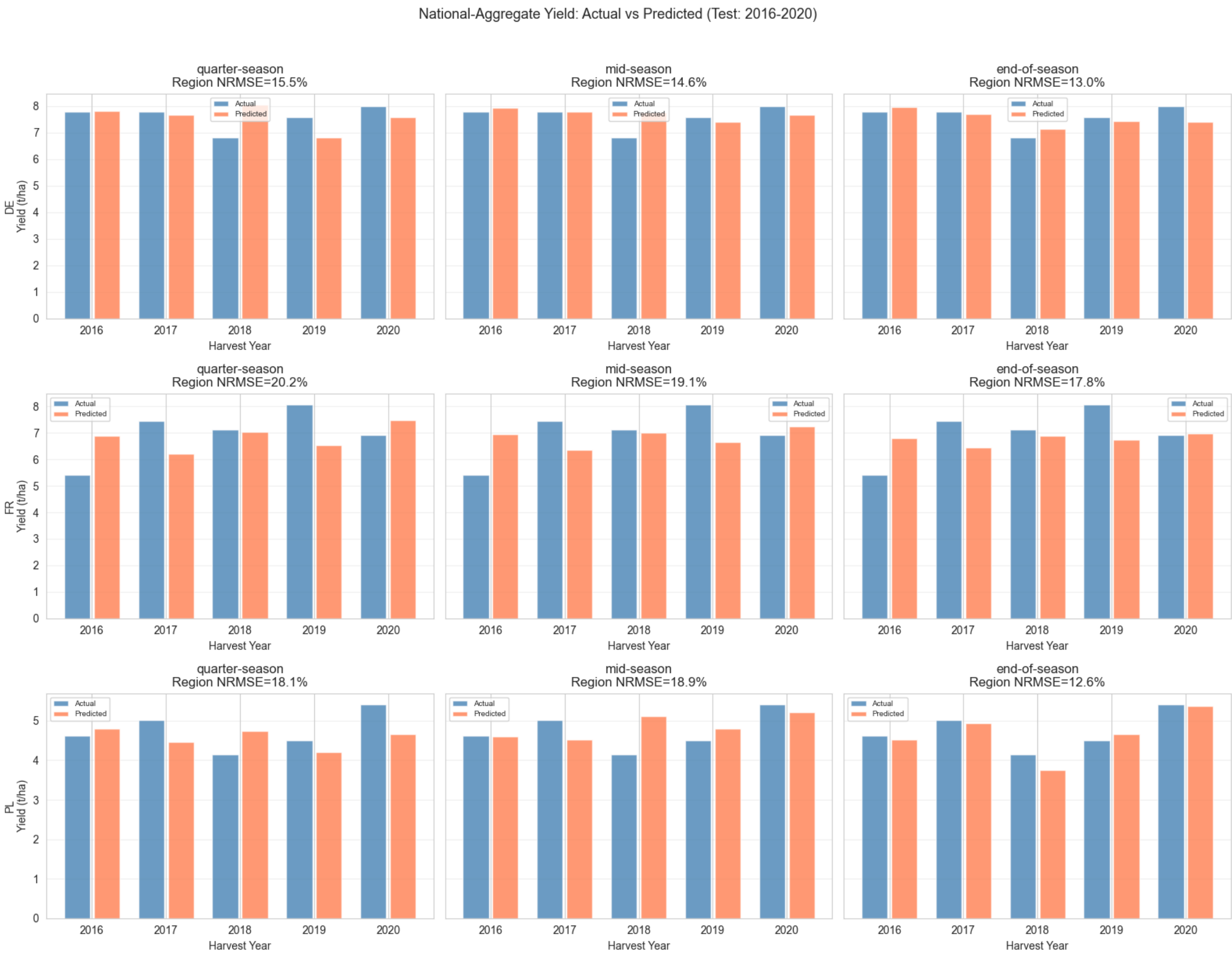

Fig. D9 NRMSE Results on the test set for France, Germany and Poland

Appendix E Yield Forecasting Accuracy

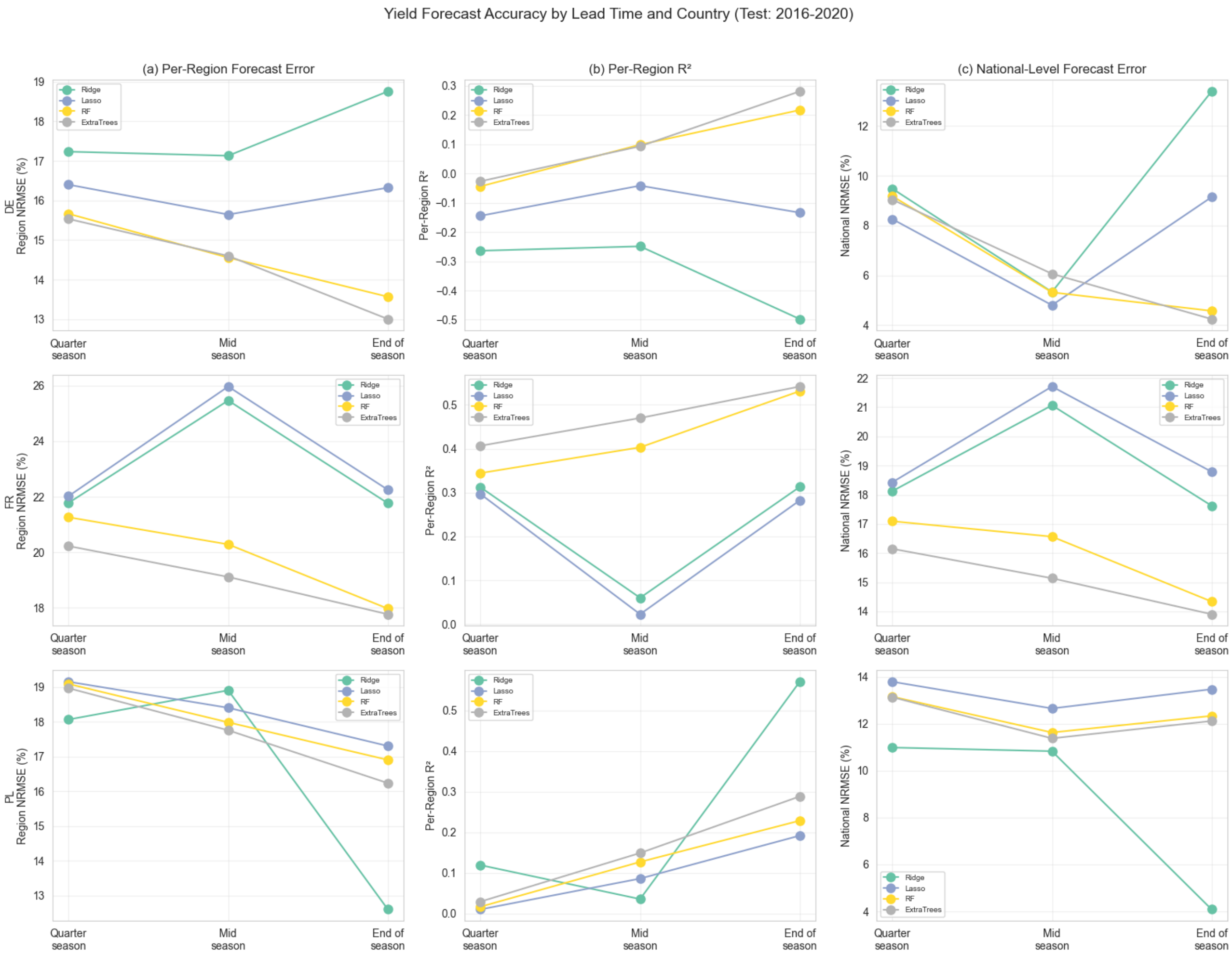

Fig. E10 Yield Forecasting Accuracy for each Model and Country